IBM Garage Case Study

Smart City Construction
Smart Water Supply Network Platform Based on Edge Computing

#IoT, #5G, #EdgeComputing, #DesignThinkingMethodology

Project Intro

Project Overview

RiFeng Enterprise Group Co., Ltd. is a renowned enterprise in China specializing in the R&D, production, and application of new plastic piping systems. As a key promoter of China's water supply pipe upgrades and a national high-tech enterprise under the Torch Plan, RiFeng regards product quality as its core competitiveness. It possesses a national-certified enterprise technology center and CNAS-certified laboratories, holding over 600 domestic and international patents.

"This MVP gives RiFeng the ability to innovate, enhance the brand competitiveness, and lay a technical foundation for opening the smart city pipe network market."

My Role

1.Opportunity Discovery Workshop
2.Design Thinking Workshop
3.UX/UI Design - Edge Computing Platform

Framing Workshop

Discover Opportunities

Research & Discovery: In recent years, RiFeng decided to expand its business into the field of smart city construction. Based on pre-interview results, RiFeng identified 5 potential directions. During the workshop, RiFeng ultimately voted for three key opportunities: **Smart Water Supply**, **Safety Protection**, and **Inventory Optimization System**.

Business Model : Smart Water Supply

Business Model : Smart Drinking Water

Business Opportunities	Description	Obstacles
★ Smart Water Supply Final decision	The system can detect water pressure and flow, and analyze water pipe leakage points through the collected data. The data can then be applied as a part of smart city construction.	• Requires government support ? • Values & Benefits
★ Safety Protective Water Supply internal service	Internal innovation services, including leakage detection, pipe certification testing, membership services, and construction information.	• How to analyze the data of 60 million members? • How to improve user experience and engagement?
Smart Drinking	Internal innovation services include leakage detection, pipe certification testing, membership services, and construction information.	• How to win customers' trust in providing clean water?
★ Inventory Optimization System	• Lowering transportation costs? • Reducing inventory?	• How to use existing data?
Wastewater System	Combine Geographic Information System (GIS) information to mark leakage points on the map.	• How to use existing data? • What are the business model and revenue model?

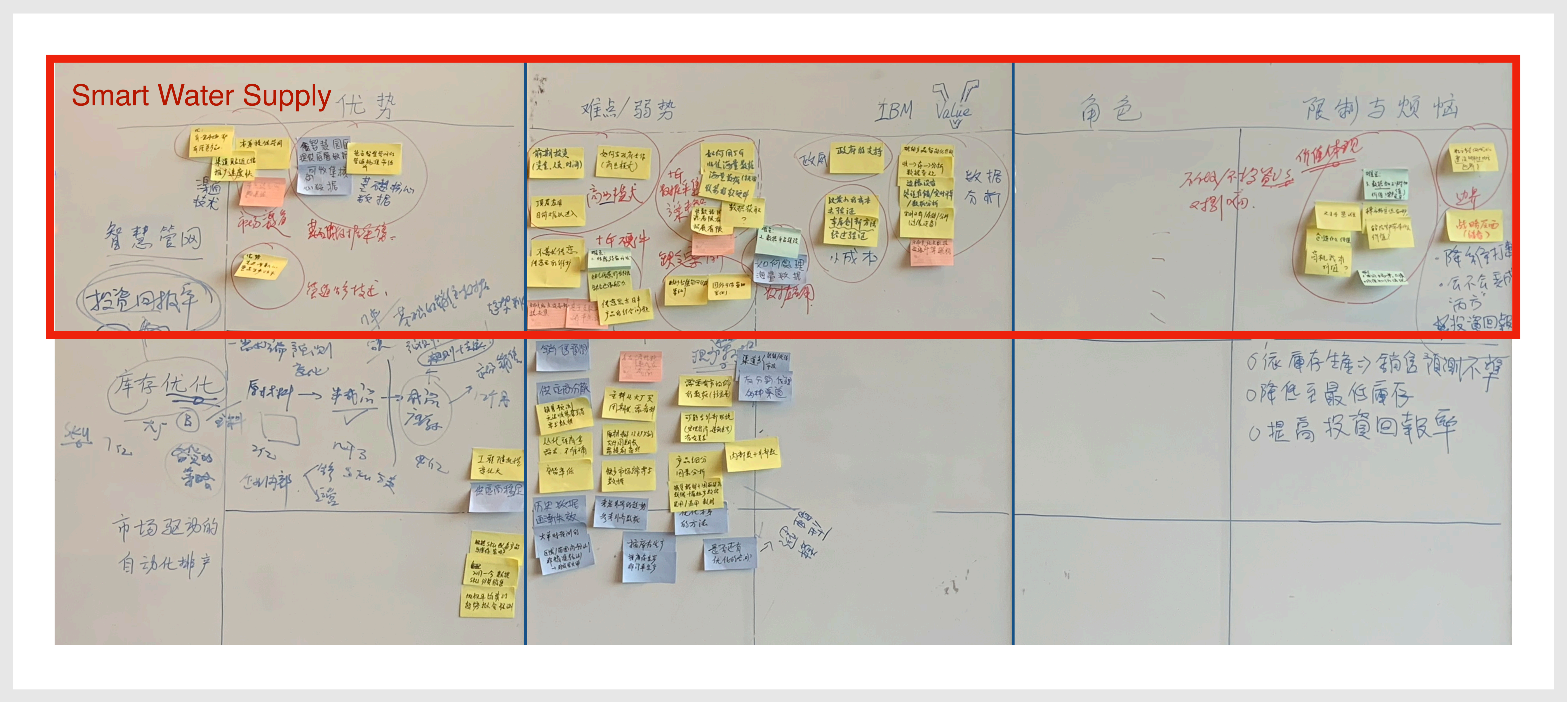
Framing Workshop

Research & Investigation

In recent years, RiFeng has decided to expand its business in the field of intelligent city construction.
Root

According to the preliminary visit results, the client and we agreed on five directions for this project.

Dayong ultimately voted for three opportunities - intelligent water supply, safety protection, and inventory optimization system - in this workshop.



Scenario Brainstorming

Rifeng Current Shortcoming

- Lack of urban construction scenarios
- How to collect data on water resources?
- Can the initial investment cost be recovered?

Advantages of IBM

- Verify the possibility of the intelligent water supply MVP with the minimum cost
- Intelligent empowerment - Clarify the application scenarios of water supply
- Intelligent water supply data analysis

Design Thinking Workshop

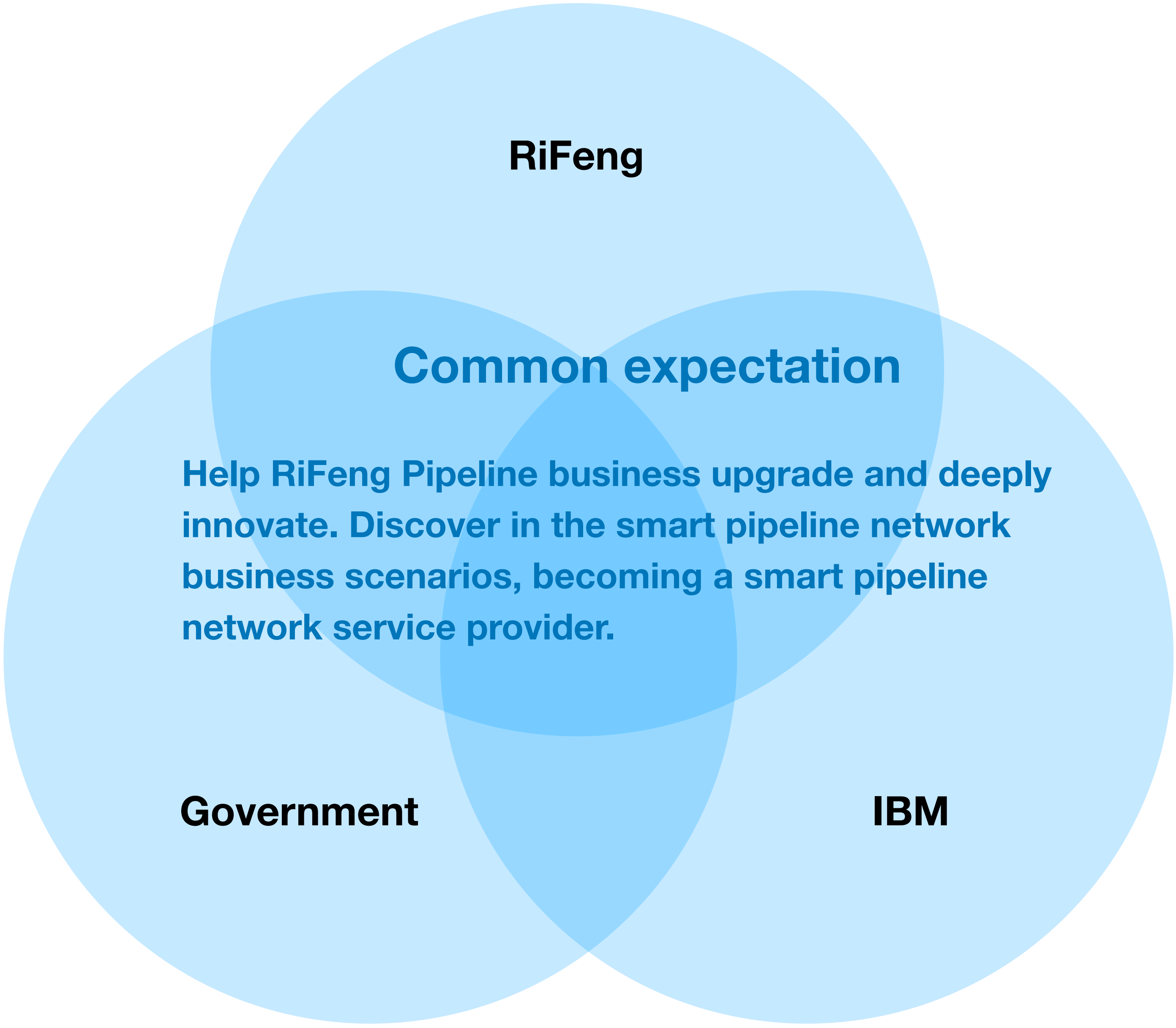
Result of Workshop

Based on the smart network business opportunities identified in the Opportunity Discovery Workshop, the overall process of the Design Thinking Workshop was designed and planned. The objective was to identify requirements for the innovation process, solutions, and final outcomes.

The workshop process primarily consisted of four steps: Stakeholder Map, Inventory of Existing Data, Analysis of Pain Points and Requirements for Key Scenarios, and Solution Ideation. Design Thinking Workshop - Stakeholder Relationship Map for Smart Pipe Network.

Smart pipe network (Final selection by the customer)
Advantages

1. Established market channels
2. Mature pipe production technology
3. Ability to collect basic and core data



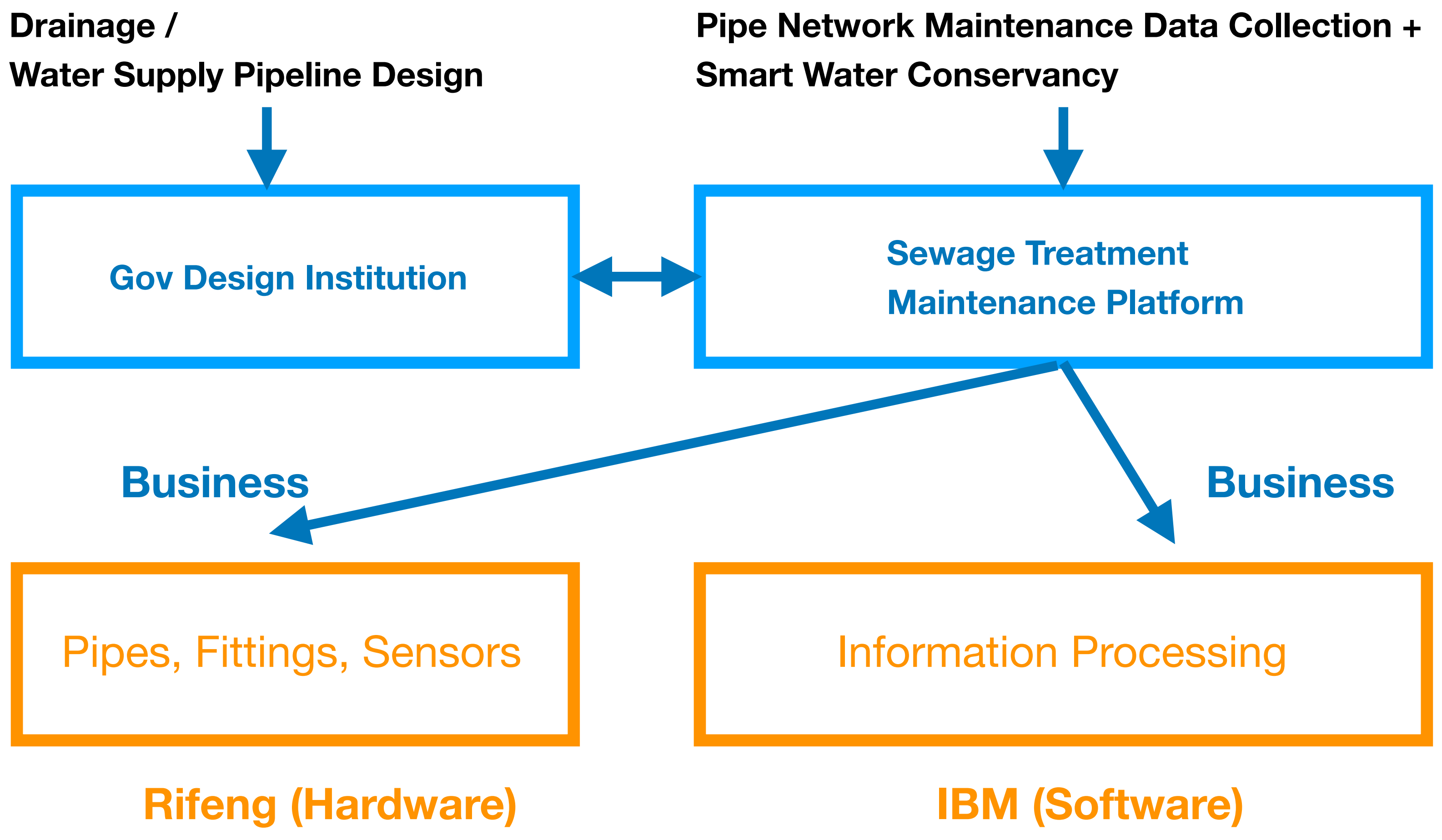
Design Thinking Workshop - Stakeholder Relationship Map for Smart Pipe Network. Smart Pipe Network Innovative Business.

Business Scenario: Smart pipe network for municipal sewage treatment plants. Main Objectives:

1. Realize data collection for sewage plant pipelines.
2. 2. Detect pipeline leakage points.

Services:

1. Provide sewage plants with real-time analysis to monitor pipeline leakage.
2. 2. Predictive maintenance of pipeline health.



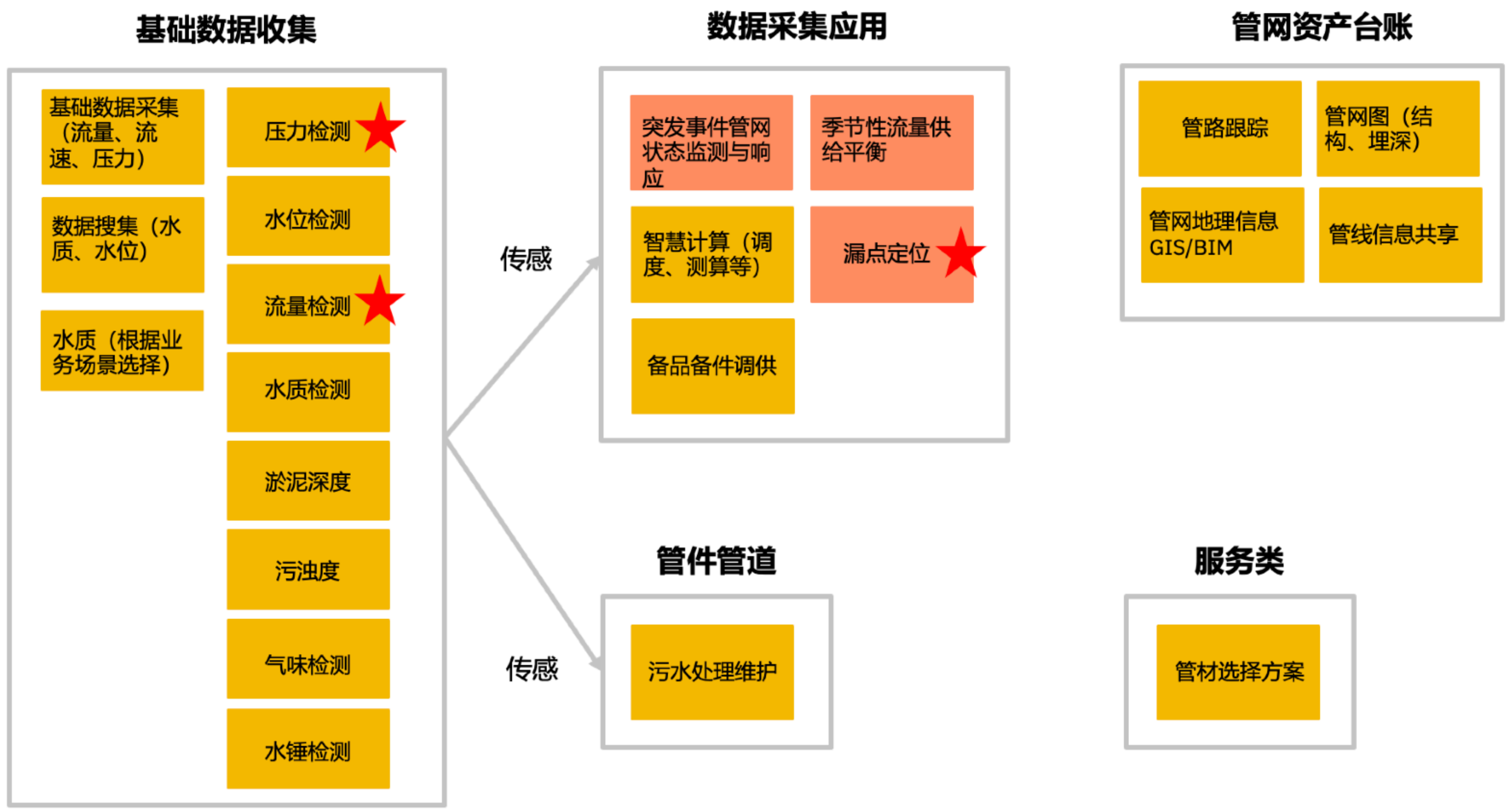
Smart pipe network business relationship map

Identify the capability requirements for RiFeng's future entry points and the capabilities provided by IBM's platform + model prediction.

Business Sorting

DTW + Business Data Analysis

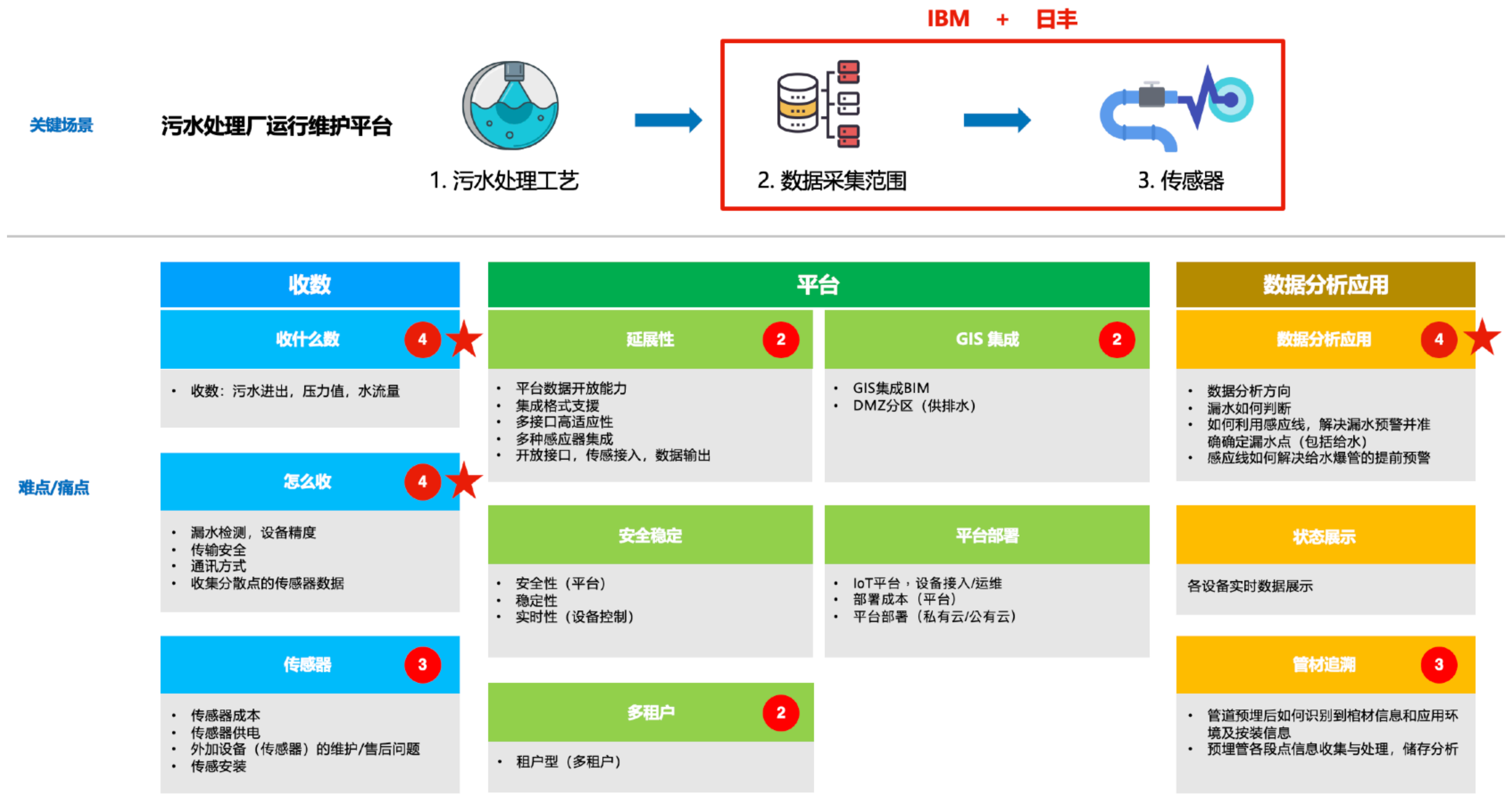
First, RiFeng colleagues were invited to identify the data required for the future **Smart Pipe Network** and categorize them during the workshop. Ultimately, the RiFeng team selected the most suitable data points and scenarios for market entry; **pressure detection**, **flow monitoring**, and **leakage localization** were chosen as the three core scenarios (or data types) to serve as the starting point for the Smart Pipe Network business.



Vision Map of RiFeng Municipal Pipe Network Platform.

DTW + Journey Map

Through the **data inventory** phase, **pipe network leak analysis** was selected as the first innovative **MVP** scenario. We initially identified the hardware and equipment required for **leakage localization analysis**, then categorized the output into three dimensions—**Data Collection** (how to collect), **Platform Capabilities**, and **Data Analytics** (what the data is and how to apply it)—to guide workshop participants toward purposeful results. Ultimately, this was integrated into a comprehensive **journey map** (as shown below) spanning from raw data to platform integration and finally to data analytics applications.

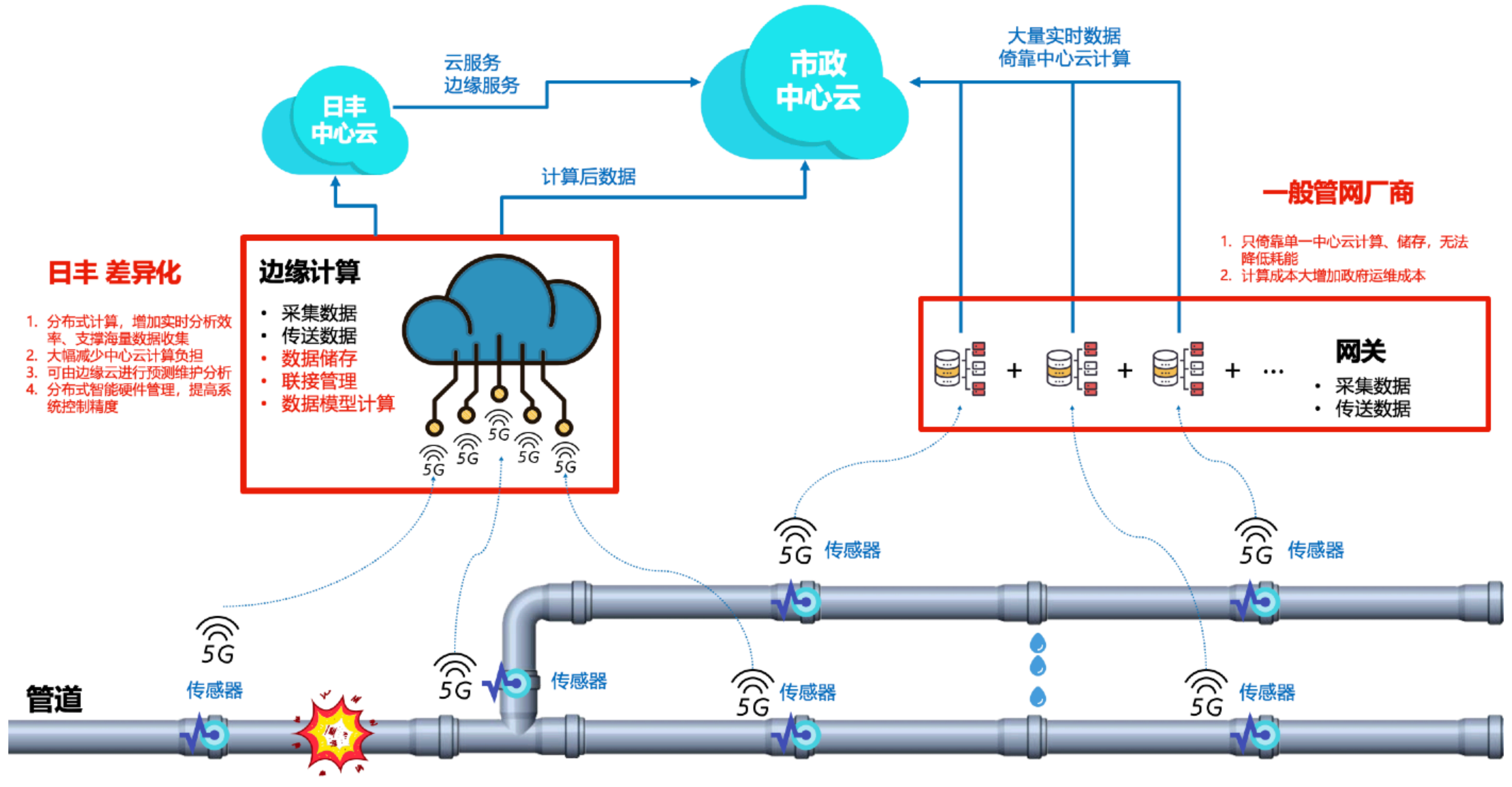


Vision Map of RiFeng Municipal Pipe Network Platform.

Ideation

DTW + Brainstorming

Based on the previous leak analysis scenario, we collaborated with the RiFeng team to ideate a future solution. By leveraging **Edge Computing and 5G capabilities** for future integration into municipal platforms, this scenario empowers RiFeng with fundamental **edge model computing** capabilities.



Leaking analysis scenario = Edge Computing + 5G tTansmission + Rifeng Private Cloud

DTW + Ideate

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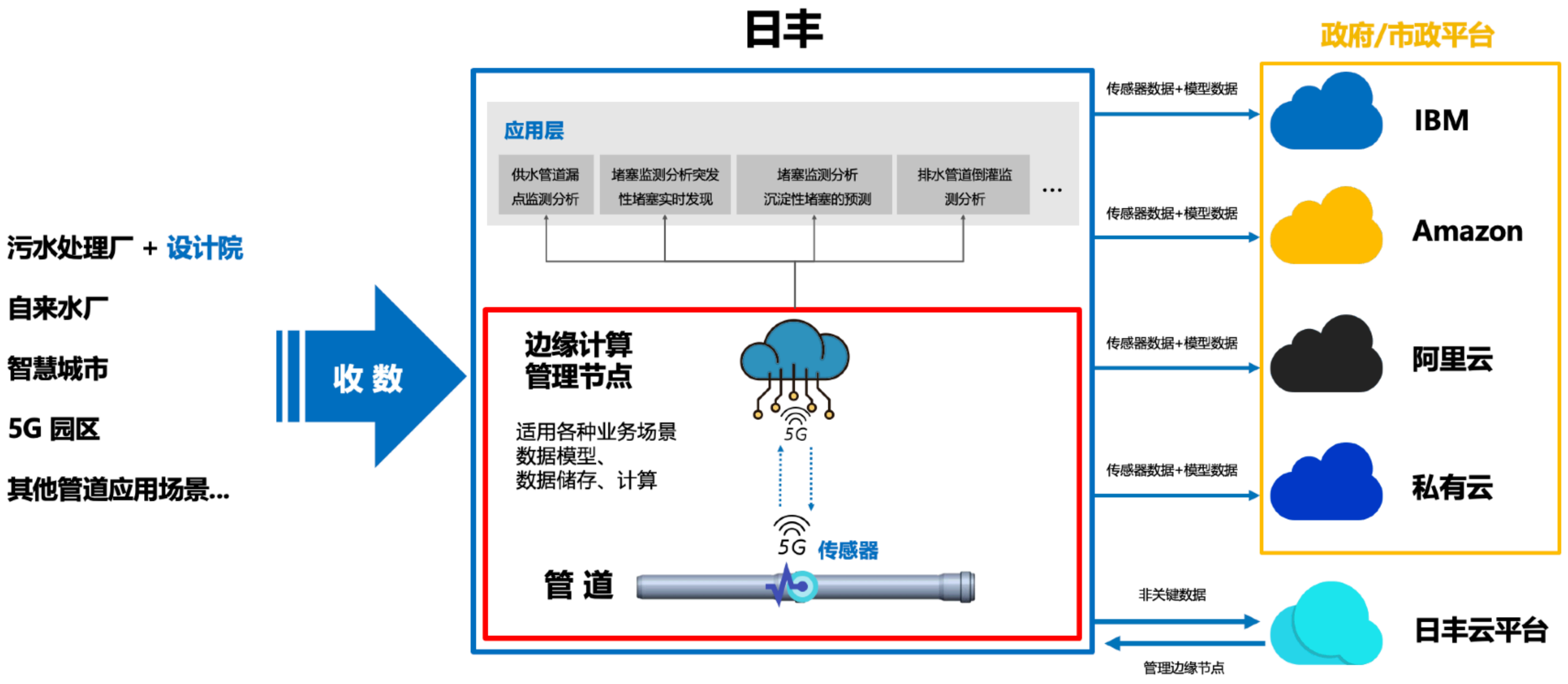


Vision Map of RiFeng Municipal Pipe Network Platform.

MVP Development -
Leakage Localization Edge System Model

Data is transmitted via **5G sensors** to an **edge node server**, where **edge computing** technology and an embedded **leak detection model** are utilized. The processed data is then sent from the edge node to a **cloud server**. Once the **cloud platform** visualizes the leak location, a **leakage alert** is successfully generated.

5G Sensor

Computing Technology

- Sensor Data Formatting
- Edge Deep Computing Model; DNN Partitioning; DNN Right-sizing

Communication Technology

- Data Transmission and Reception between Sensors
- Intermittent Communication

Edge Services

Computing Technology

- Interfacing with Edge Partitioned Model Outputs
- Full Deep Learning Models
- Collaborative Computing Model for Sensor Data within Local Networks

Communication Technology

- Edge Sensor Data Collection
- Local Network Signal Chain Establishment
- Intermittent Computing for Local Sensors

Cloud Sensor

Computing Technology

- Establishment of Full Pipe Network Sensor Depth Maps
- Generalizable Performance of Leakage Prediction across Multiple Environmental Nodes
- Pattern Schema/Graph for Urban Water Supply Networks

Communication Technology

- Edge Sensor Data Collection
- Regional Sensor Data Acquisition

MVP

MVP Iteration -
Leak Detection Model Evolution

Data is transmitted via **5G sensors** to an **edge node server**, where **edge computing** technology and an embedded **leak detection model** are utilized. The processed data is then sent from the edge node to a **cloud server**. Once the **cloud platform** visualizes the leak location, a **leakage alert** is successfully generated.

First Iteration

Accuracy **0.9943**

Applicable Scenarios

- Leakage Point 1 at any opening angle;
- Leakage Point 2 at any opening angle;
- Leakage Points 1 & 2 both at any opening angle simultaneously.

Constraints/Issues

Model accuracy is suspiciously high, indicating potential overfitting issues.

Second Iteration

Accuracy **0.9925**

Applicable Scenarios

- Leakage Point 1 at any opening angle;
- Leakage Point 2 at any opening angle;
- Leakage Points 1 & 2 both at any opening angle simultaneously.

Third Iteration

Accuracy **0.9601**

Applicable Scenarios

- Leakage Point 1 at any opening angle;
- Leakage Point 2 at any opening angle;
- Leakage Points 1 & 2 both at any opening angle simultaneously.

Constraints/Issues

The model's leak detection boundary is unstable

Fourth Iteration

Accuracy **0.9307**

Applicable Scenarios

- Leakage Point 1 at any opening angle;
- Leakage Point 2 at any opening angle;
- Leakage Points 1 & 2 both at any opening angle simultaneously.

Delivered Model

- Corrected the model overfitting issue;
- Model performance has stabilized.

MVP

MVP Development - Smart Pipe Network Visual Management Platform

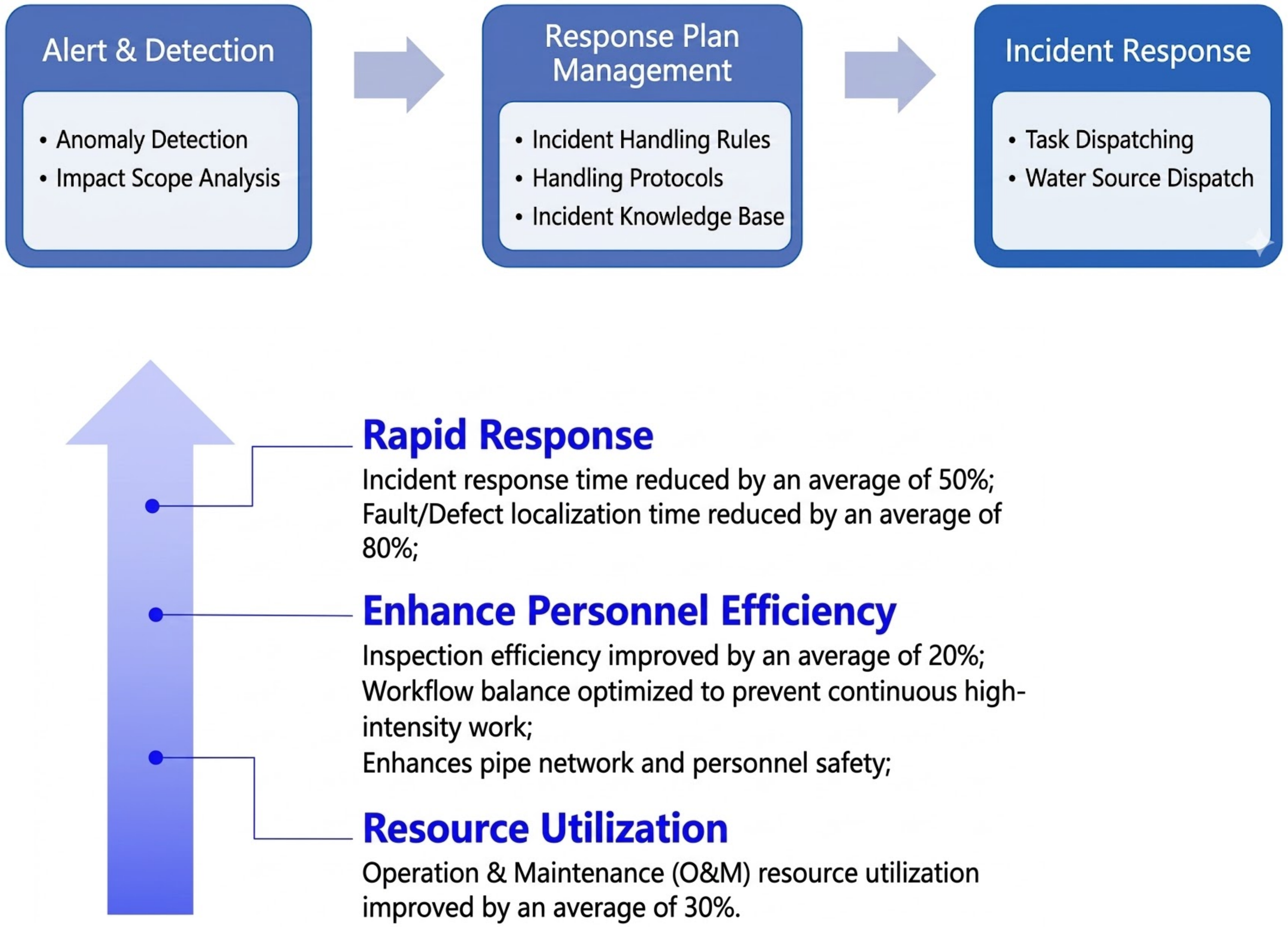
Once the management platform receives the processed data from the edge nodes, it can directly display the node status and quickly locate leakage points, achieving the objective of early warning for leak detection.

RiFeng Municipal Pipe Network Platform - Pipe Network Leakage Monitoring Map

MVP Vision

In the construction of **Smart Pipe Networks**, a rapid response to emergencies is the cornerstone of providing more efficient decision-making

In recent years, frequent accidents involving damage to water supply pipelines and facilities caused by construction have posed serious safety risks and impacted water supply security. By leveraging a **Pipeline Leak Detection Platform**, these concerns can be addressed, enabling **disaster early warning** and a significant reduction in **water resource waste**.



Benefits of Implementing a Smart Pipe Network Platform (Leak Detection)